THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Light – Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** −10 cm	$f = R/2$
2	**(a)** +2 D	$P = 1/f(\text{m}) = 1/0.5$
3	**(a)** 1.5	$n = c/v$
4	**(a)** Correct	For all positions
5	**(b)** Student A is correct	Wider field of view
6	**(c)** at C	Image at C
7	**(a)** goes undeviated	No deviation
8	**(c)** towards the normal	Denser medium
9	**(c)** dioptre	m^{-1}
10	**(c)** convex, object between F and O	Magnifying glass

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(i) angle of incidence = angle of reflection (ii) the incident ray, the reflected ray and the normal lie in the same plane.	1+1
12	Point on the principal axis where rays parallel to it meet after reflection.	2
13	Due to refraction of light at the water–air surface.	2
14	False — a convex lens is used; a concave lens always gives diminished images.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$u = -20$, $f = -15$: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60}$ ⇒ $v = -60$ cm; $m = -\frac{v}{u} = -3$; real, inverted, magnified	1+1+1
16	$\frac{\sin i}{\sin r}$ = constant for a given pair of media; $n_{ag} = 1/1.5 = 0.67$	2+1
17	(i) image beyond C, real, inverted, magnified (ii) image behind the mirror, virtual, erect, magnified	1½+1½
	OR: Torches/headlights (reflectors), shaving mirrors, dentist's mirrors, solar furnaces	3
18	Ability to converge/diverge light, $P = 1/f$ (m); $P = 1/(-0.25) = -4$ D	1+2
	OR: (i) ratio of speed of light in vacuum to that in medium (ii) central point of a lens through which light passes undeviated (iii) line through centres of curvature/optical centre	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$ ⇒ $v = +30$ cm; $h' = mh = -8$ cm; real, inverted, magnified (ii) object between F and 2F, image beyond 2F (iii) $m = v/u = -2$	3+1+1
	OR: (i) (a) at F, real, inverted, highly diminished (b) between F and 2F, real, inverted, diminished (c) at 2F, real, inverted, same size (ii) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	3+2
20	(i) the two refracting surfaces are parallel; the bending at both surfaces is equal and opposite (ii) lateral displacement (iii) decreases (iv) $v = c/n = 2 \times 10^8$ m/s	2+1+1+1
	OR: (i) object on left, distances from pole; along incident light +, opposite −; heights above axis +, below − (ii) $f = R/2 = +16$ cm (iii) wide field of view; erect image	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **Light – Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** infinite	Flat surface
2	**(a)** concave lens of focal length 40 cm	$f = 1/P = -0.4$ m
3	**(c)** +1	Same size, erect
4	**(c)** Incorrect — it is optically the densest; light is slowest	$v = c/n$
5	**(b)** Student A is correct	Parallel rays
6	**(b)** concave	Magnified image
7	**(d)** virtual, erect and diminished	All positions
8	**(b)** positive and less than 1	Diminished, erect
9	**(d)** 2.25×10^8 m/s	$3/1.33$
10	**(c)** becomes parallel to the principal axis	Reversibility

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Radius of the sphere of which the mirror is a part; $R = 2f$.	2
12	Power of a lens of focal length 1 metre.	2
13	Speed of light changes in different media.	2
14	False — a convex mirror always forms a diminished, virtual, erect image.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$u = -10$, $f = +15$: $\frac{1}{v} = \frac{1}{15} + \frac{1}{10} = \frac{1}{6} \Rightarrow v = +6$ cm; $m = -v/u = 0.6$; virtual, erect, diminished	2+1
16	$u = -30$, $f = -15$: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = -\frac{1}{15} - \frac{1}{30} = -\frac{1}{10} \Rightarrow v = -10$ cm; $m = v/u = \frac{1}{3}$; virtual, erect, diminished	2+1
17	(i) at F, virtual, erect, highly diminished (ii) between F and O, virtual, erect, diminished	1½+1½
	OR: Convex mirror: rear-view mirror, street light reflectors; convex lens: magnifying glass, camera, spectacles for hypermetropia	3
18	Fastest in water (lowest n); densest: turpentine (highest n)	1½+1½
	OR: $R = 2f$; $R = -24$ cm	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) (a) between F and C, real, inverted, diminished (b) at infinity, real, highly magnified (c) behind mirror, virtual, erect, magnified (ii) between P and F	4+1
	OR: (ii) $\frac{1}{v} = -\frac{1}{10} + \frac{1}{25} = -\frac{3}{50} \Rightarrow v = -16.7$ cm; $m = -v/u = -0.67 \Rightarrow h' = -3.3$ cm (iii) real, inverted, diminished, between F and C	1+3+1
20	(i) converging (convex) (ii) $f = 1/2 = 0.5$ m = 50 cm (iii) $\frac{1}{v} = \frac{1}{50} - \frac{1}{25} = -\frac{1}{50} \Rightarrow v = -50$ cm (iv) virtual, erect, magnified ($m = 2$)	1+1+2+1
	OR: (i) bending of light at the boundary; incident ray, refracted ray, normal in one plane; Snell's law (ii) $n = c/v$ (iii) light from coin bends away from normal at water–air surface; appears to come from a higher point	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.